Winning the Space Race with Data Science

From raw launch records to geographical intelligence and predictive modelling

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IBM Data Science Professional Certificate • Capstone presentation

github.com/eokorouka-jpg/SpaceX-Falcon9-Data-Science-Capstone

Executive summary

90
labelled ML launches

56
Folium / Dash records

83.33%
best held-out accuracy

- End-to-end workflow: collection → cleaning → SQL / EDA → mapping → dashboard → classification
- Landing success varies across launch sites, payload bands and booster versions in the sampled data
- Logistic Regression, SVM and KNN tied on the 18-flight held-out test set

Business question & analytical objectives

Can historical launch characteristics help explain and predict successful Falcon 9 first-stage landings?

- Explore landing outcomes by site, orbit, flight history and payload
- Investigate geographical context and transportation access
- Create interactive tools for filtering and inspecting launches
- Compare tuned classifiers and evaluate prediction errors

Methodology: one connected workflow

01

COLLECT

SpaceX API + Wikipedia

02

PREPARE

clean, label, encode

03

EXPLORE

Python EDA + SQLite

04

VISUALISE

Folium + Plotly Dash

05

PREDICT

4 tuned classifiers

Python • pandas • SQL • Folium • Plotly Dash • scikit-learn

Data collection: two complementary sources

SpaceX REST API

- Request historical launch records
- Expand rocket, core, payload and pad IDs
- Flatten JSON to a structured table

Wikipedia launch tables

- Inspect HTML tables with BeautifulSoup
- Extract booster, launch and mission fields
- One saved attempt returned HTTP 403

Source note: an HTTP 403 attempt is not presented as a successful scrape.

Data wrangling & target definition

90 × 18
labelled ML dataset

90 × 80
encoded feature matrix

0 / 1
binary landing class

- Filter Falcon 9 launches; standardise fields and address missing payload mass
- Class = 1: successful first-stage landing; Class = 0: other landing outcomes
- One-hot encode categorical attributes; align feature rows with target labels
- Standardise numeric features before fitting classification models

Why the dataset sizes differ

Analytical component	Records	Purpose
API / ML labels	90	Model training and evaluation
SQL launch table	101	Structured launch-history queries
Folium + Dash	56	Interactive geographic and dashboard analysis

These are different lab extracts and should not be combined into a single denominator.

Exploratory analysis: launch-site success



KSC LC-39A: 10 successful landings / 13 launches — strongest observed site in the 56-row sample.

EDA result: launch volume & landing outcomes

Launch-site label	Launches	Successful	Success rate
CCAFS LC-40	26	7	26.92%
CCAFS SLC-40	7	3	42.86%
KSC LC-39A	13	10	76.92%
VAFB SLC-4E	10	4	40.00%
TOTAL	56	24	42.86%

Data quality: CCAFS LC-40 and CCAFS SLC-40 are near-duplicate location labels; treat carefully.

EDA: payload mass & booster generations



Highest observed payload-band success: 2,000–4,000 kg (13 / 21 launches).

Small groups and historical confounding limit causal interpretation.

SQL analysis: queries answered with evidence

101
launch records in SQL

45,596 kg
NASA CRS total payload

2,928.4 kg
F9 v1.1 average payload

- Earliest successful ground-pad landing: 22 December 2015
- SQL distinguished mission success from first-stage landing outcomes
- Date text was converted from DD-MM-YYYY to ISO order for filtering

SQL output: landing outcomes, 2010–2017

Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Period: 4 Jun 2010 – 20 Mar 2017 • 31 records

SQL findings: booster and landing history

Drone-ship landing attempts

5 / 10

successful in the 2010–2017 filtered period

Maximum-payload booster IDs

12

distinct IDs; all labelled F9 B5 in the SQL results

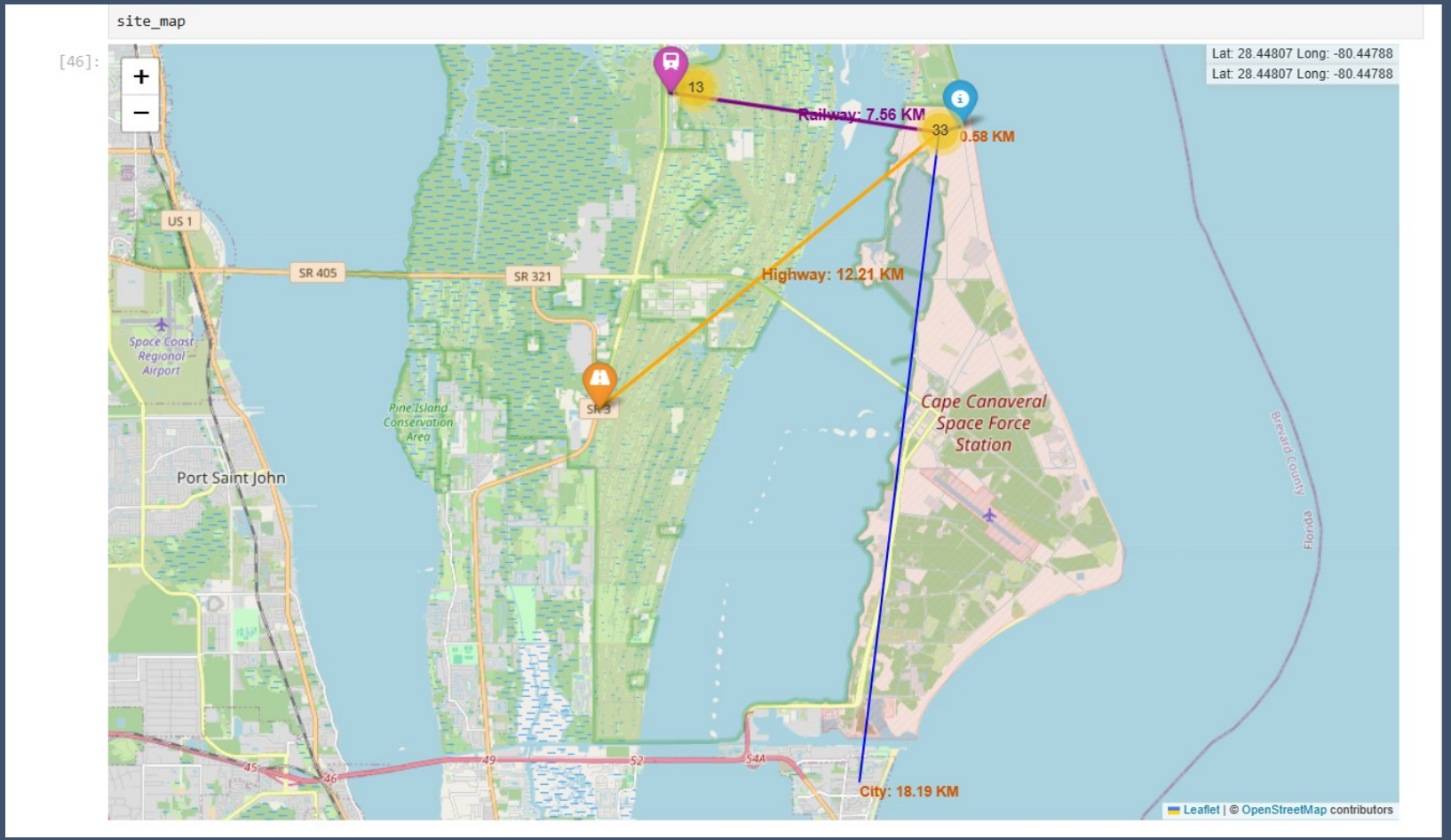
Two failed drone-ship landings in 2015 were identified in January and April.

Folium: interactive geographical exploration

- Launch-site coordinates plotted with circles and labels
- Green / red landing-outcome markers and MarkerCluster for exploration
- MousePosition and Haversine straight-line distances
- Lines connect LC-40 to selected coast, rail, highway and city points

Map evidence: screenshot captured from the completed Jupyter Folium notebook.

Folium results: LC-40 proximity map



Measured to selected points, not verified nearest features • Coast 0.58 km • Rail 7.56 km • Highway 12.21 km • City 18.19 km

Geographic interpretation & limitations

Selected geographical point	Distance from LC-40
Coastline	0.58 km
NASA railway point	7.56 km
Florida SR 3 highway point	12.21 km
City reference point	18.19 km

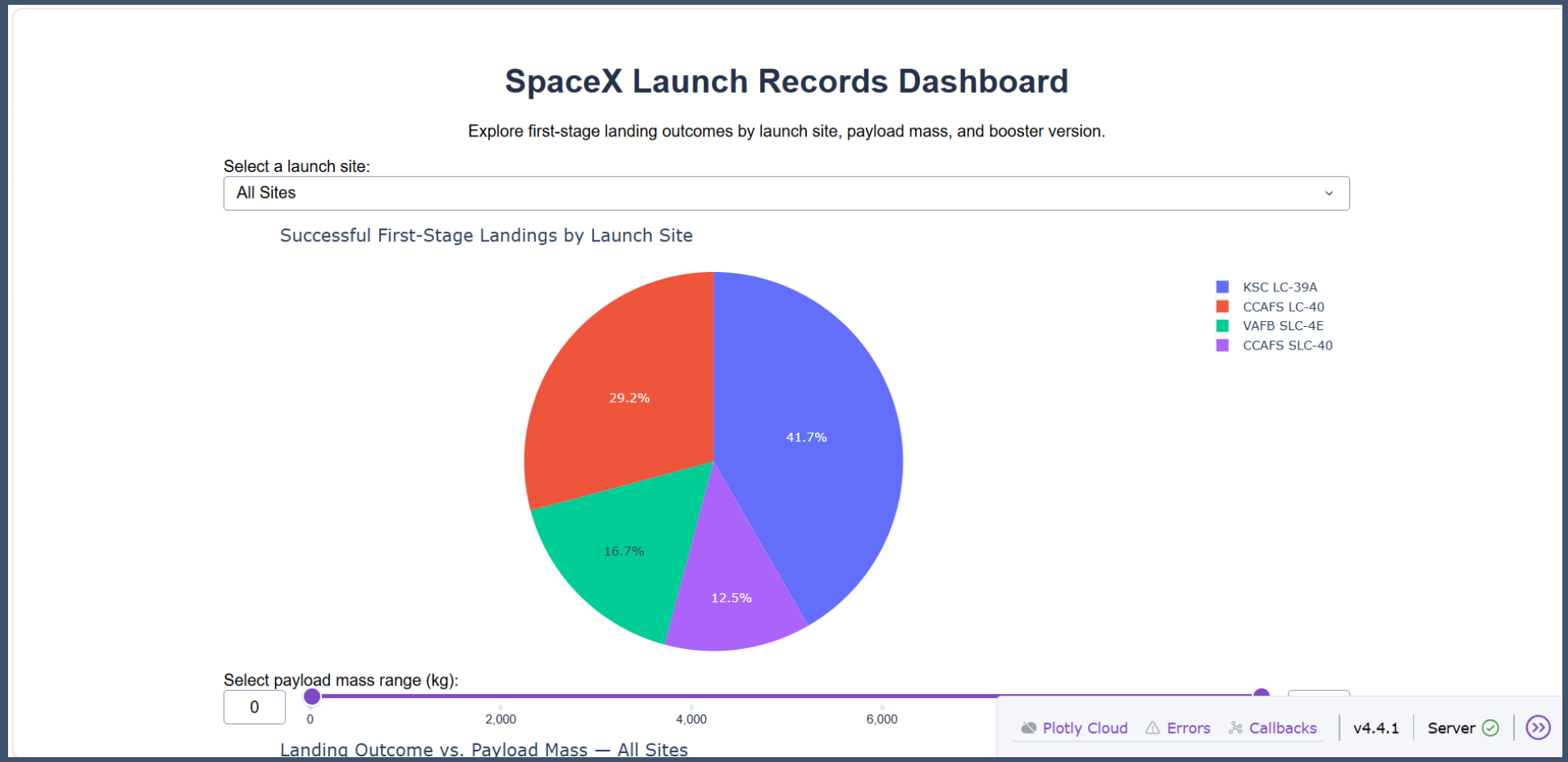
Coastal placement may facilitate trajectories over water; transport links may support logistics. These distances do not prove site-selection causes or nearest-feature proximity.

Plotly Dash: an interactive analytics app

- Launch-site dropdown: all sites or a selected location
- Pie chart: successful versus unsuccessful landings
- Payload-range slider: interactively filter launches
- Scatter plot: payload mass × outcome, coloured by booster version

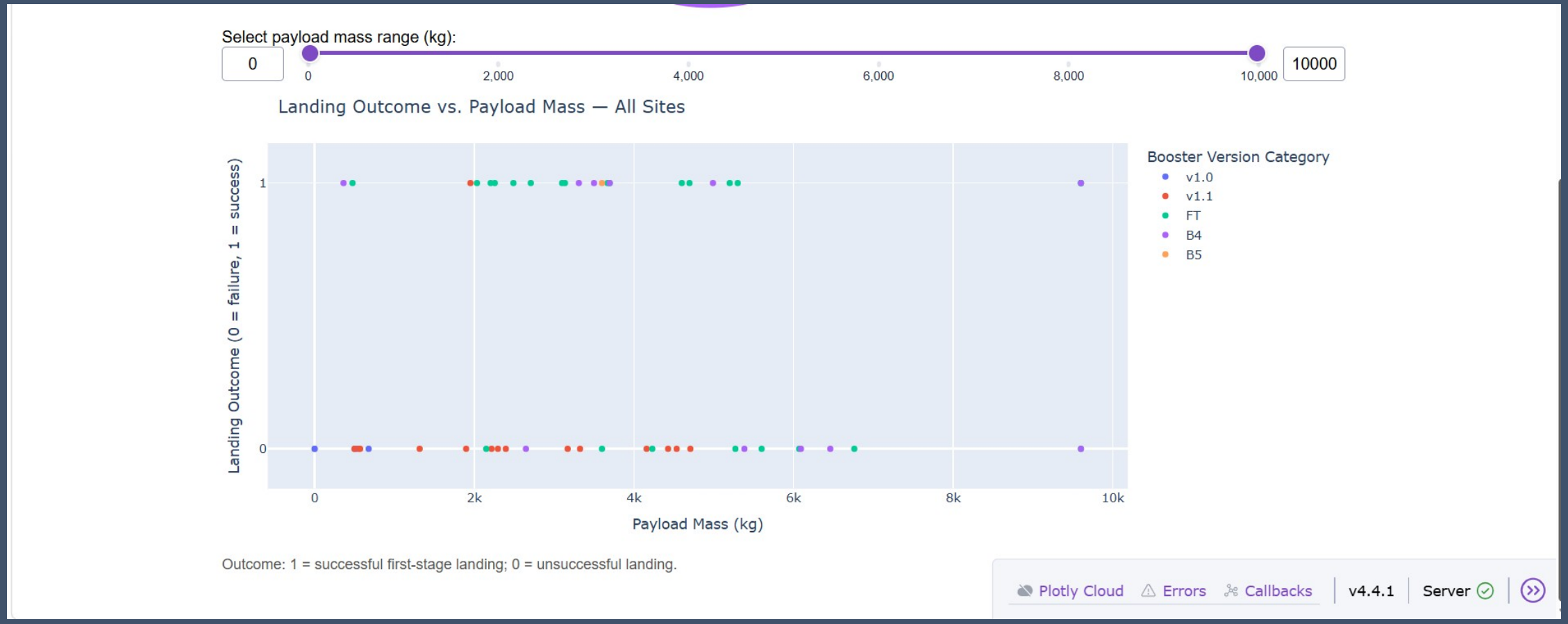
Working locally at http://127.0.0.1:8050/ • source code included in GitHub repository

Dashboard result | successful landings by site



Of 24 successful landings, KSC LC-39A contributes 10 (41.7%). Its site-specific success rate is 10/13 = 76.92%.

Dashboard result | payload and booster version



Interactive scatter: each point represents a launch; 1 = successful landing, 0 = unsuccessful. Colours distinguish booster versions.

Dashboard insights: what changed with filters

76.92%
KSC LC-39A success

61.90%
2–4 tonne payload band

66.67%
FT booster version

- KSC LC-39A: 10 successful of 13 sampled launches
- 2,000–4,000 kg: 13 successful of 21 launches
- FT boosters: 16 successful of 24 launches; B5 had only one record

Machine learning: modelling pipeline

01

FEATURES

80 encoded predictors

02

SPLIT

80% train / 20% test

03

TUNE

GridSearchCV; 10 folds

04

COMPARE

LR, SVM, Tree, KNN

Evaluation: 18 held-out launches; accuracy plus confusion matrices.

Classification results: held-out accuracy



Three classifiers tied at 83.33% (15 correct out of 18). No unique winner on this test split.

Confusion matrices: what models missed

Model	TN	FP	FN	TP	Test accuracy
Logistic Regression	3	3	0	12	83.33%
SVM	3	3	0	12	83.33%
Decision Tree	3	3	1	11	77.78%
KNN	3	3	0	12	83.33%

For the tied models, all 12 actual positive cases were detected, but three negative cases were classified positive.

Key insights: evidence, not assumptions

- Landing success differs substantially by launch-site label in the 56-launch sample
- Payload and booster-generation patterns vary, but group sizes and time trends matter
- LC-40 is coastal; selected road and railway points are within ~12 km straight-line distance
- Model test accuracy reached 83.33%, but the hold-out contains only 18 launches
- Observed associations do not establish that geography or payload caused landing outcomes

Conclusions & practical next steps

- Delivered a reproducible capstone spanning acquisition, SQL, EDA, mapping, Dash and ML
- Use the dashboard and Folium map to explore patterns rather than infer causality
- Validate on later launch periods and larger datasets; examine class imbalance and calibration
- Standardise launch-site names and verify nearest geographic features before stronger claims
- Retrain with leakage-safe preprocessing inside cross-validation pipelines

Portfolio, sources & reproducibility

GitHub repository

github.com/eokorouka-jpg/SpaceX-Falcon9-Data-Science-Capstone

- Notebooks: API, scraping, SQL, Python EDA, Folium and classification
- Dash application and supporting CSV datasets
- Interactive map: spacex_folium_final_map.html (save and commit if available)
- Presentation figures are from the project data; Folium map is a notebook screenshot

Source context: IBM Data Science Professional Certificate capstone labs; SpaceX API and Wikipedia launch data.